VISHWAKARMA
AWARDS
A Maker Bhavan Foundation Initiative

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VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID
VK26-IITI-HT-0240

Problem Statement

Project Overview

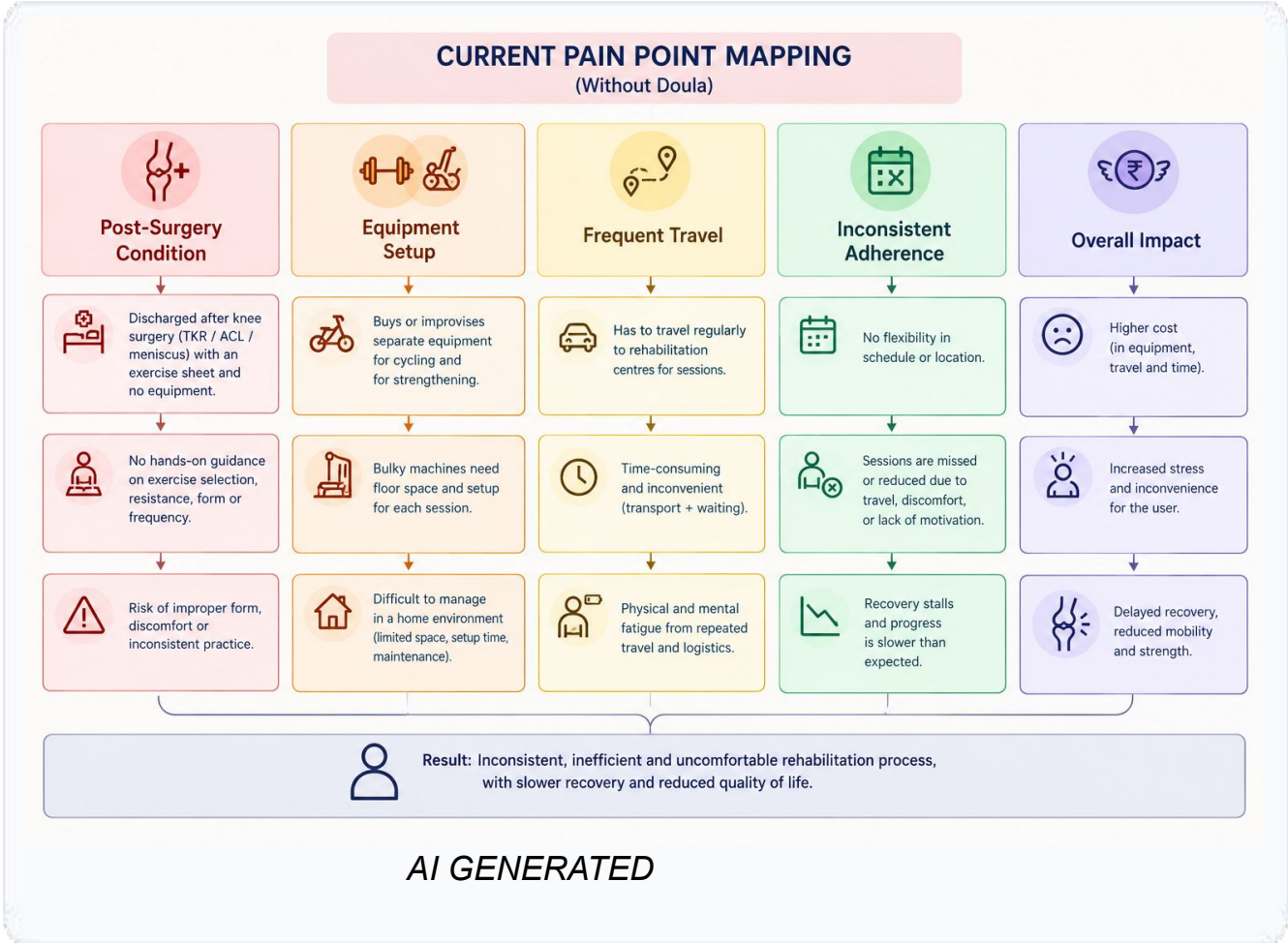
- Team ID: VK26-IITI-HT-0240
- Track: Health Tech
- Team: Gauravi Tiwari, Vedansh Tyagi, Aman Modi, Kanishk Goel, Mayank Vediya, IIT Mandi
- Mentors: Dr. Gajendra Singh [IIT MANDI], Dia Bhandari [NID ASSAM]
- **Core innovation:** One compact device that combines cycling and flexion/extension in a single unit, with mode switching by repositioning rather than a mechanical latch.

Problem Statement

- Recovery after knee procedures (TKR, ACL reconstruction, meniscus repair) needs months of daily exercise across two motion types: cycling for mobility, then flexion/extension for strengthening.
- Today these need separate, bulky equipment and repeated clinic visits. At home, patients train without guidance on resistance, form or frequency, so adherence and recovery suffer.
- **Target beneficiaries:** Older adults recovering at home in apartment-scale Indian households, often with the device bought by an adult child. A secondary user is the physiotherapist who monitors progress.

User Journey

Current Pain Point Mapping



AI GENERATED

Before State (without Doula)

- Discharged after knee surgery (TKR / ACL / meniscus) with an exercise sheet and no equipment
- Buys or improvises separate equipment for cycling and for strengthening
- Bulky machines need floor space and setup for each session
- No guidance on resistance, form or frequency, so exercise is improper or inconsistent
- Frequent travel to rehabilitation centres; adherence drops and recovery stalls

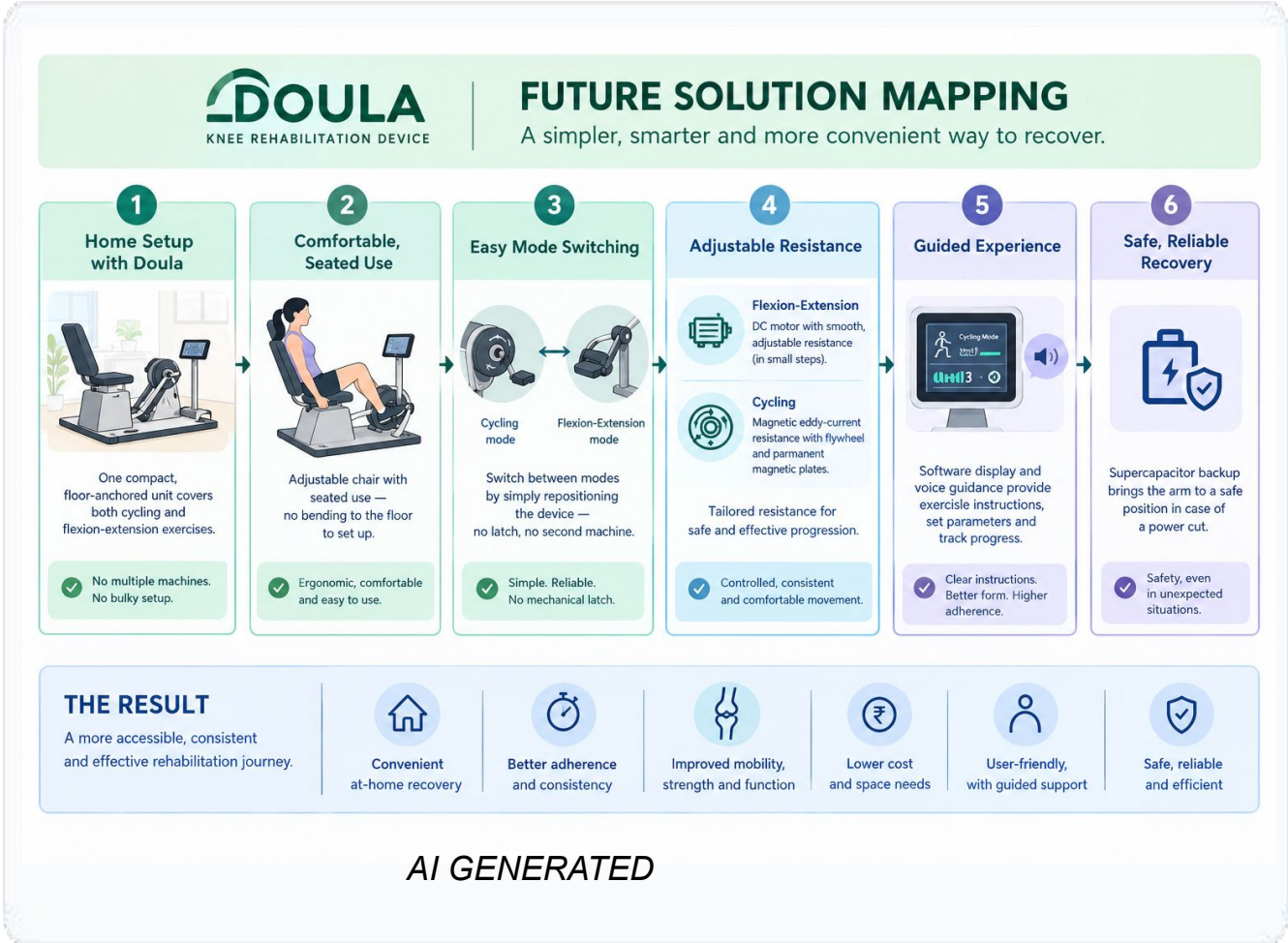
Quantified Efficiency

Equipment: 2 separate machines → 1 integrated unit

- Cost: separate machines ≈ ₹28,000-40,000 vs Doula target ≈ ₹15000
- Floor space: ≈ 1.2 m² vs Doula target ≈ 0.4 m²

User Journey

Future Solution Mapping

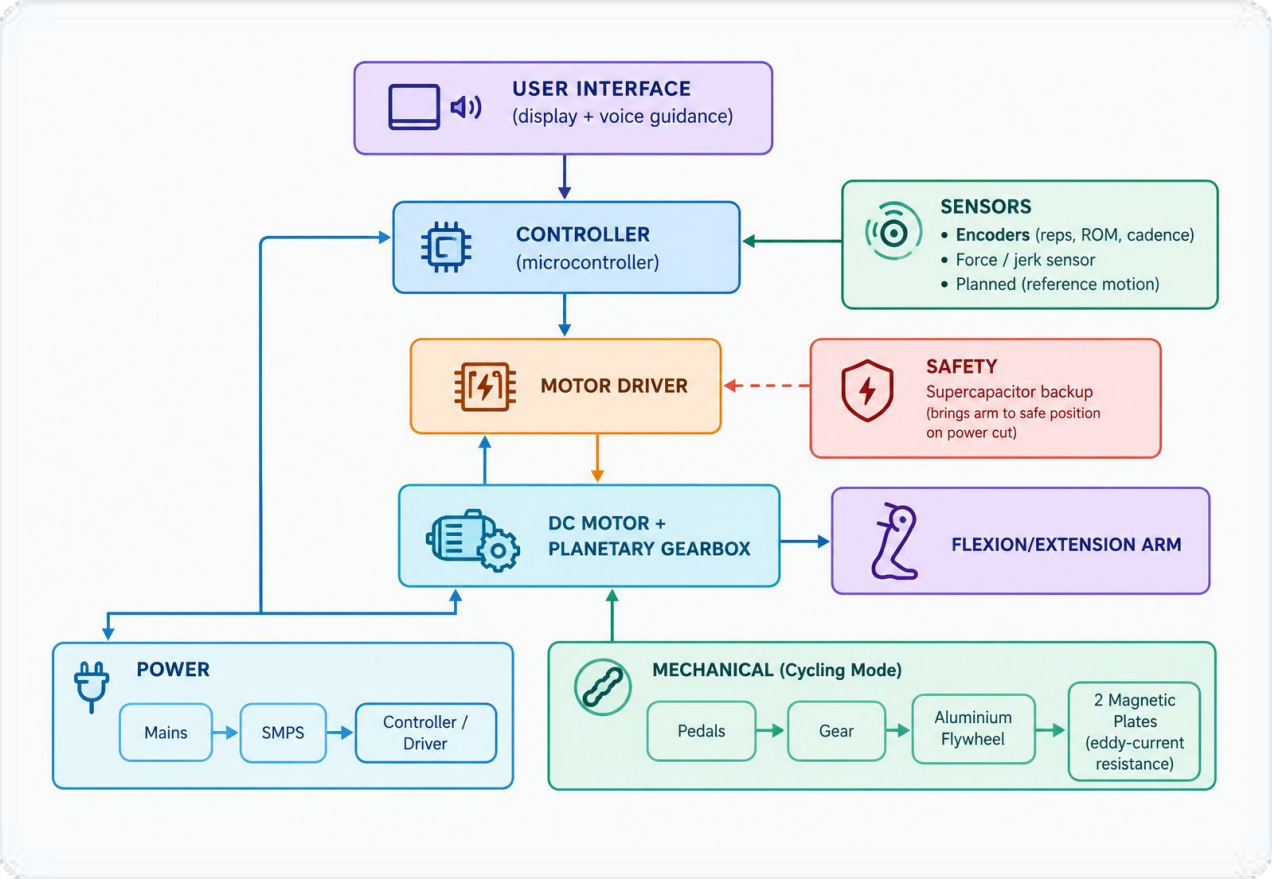


AI GENERATED

After State (with Doula)

- One compact, floor-anchored unit covers cycling and flexion/extension
- Seated use with an adjustable chair, with no bending to the floor to set up
- Mode switched by repositioning the device, with no latch and no second machine
- Adjustable resistance in small steps: DC motor (flexion/extension) and magnetic eddy-current (cycling)
- Software display and voice guidance for exercise instructions
- Supercapacitor backup brings the arm to a safe position on a power cut

System Architecture & Specs



Hardware Selection

- Microcontroller: [ESP32]
- Motor: DC geared motor with planetary gearbox
- Motor driver: Rhino MD20 Amp
- Sensors: rotary encoders (ROM, reps, cadence); load-cell force sensor for spasm detection *(planned)*
- Communication: UART / I2C between controller and display
- Safety: supercapacitor module

Architecture Rationale: Mode switching by repositioning instead of a mechanical latch was chosen to reduce part count and complexity and improve reliability.

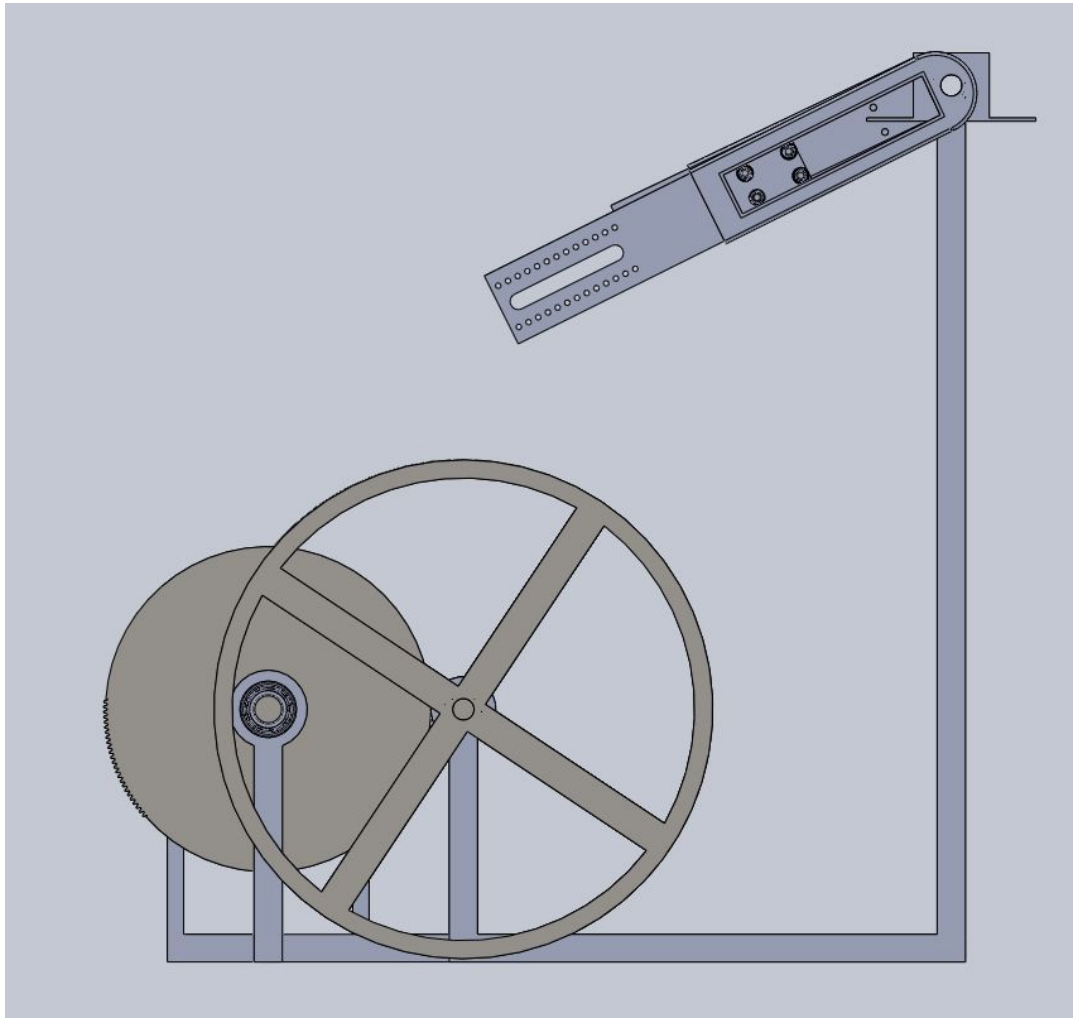
System Architecture & Specs

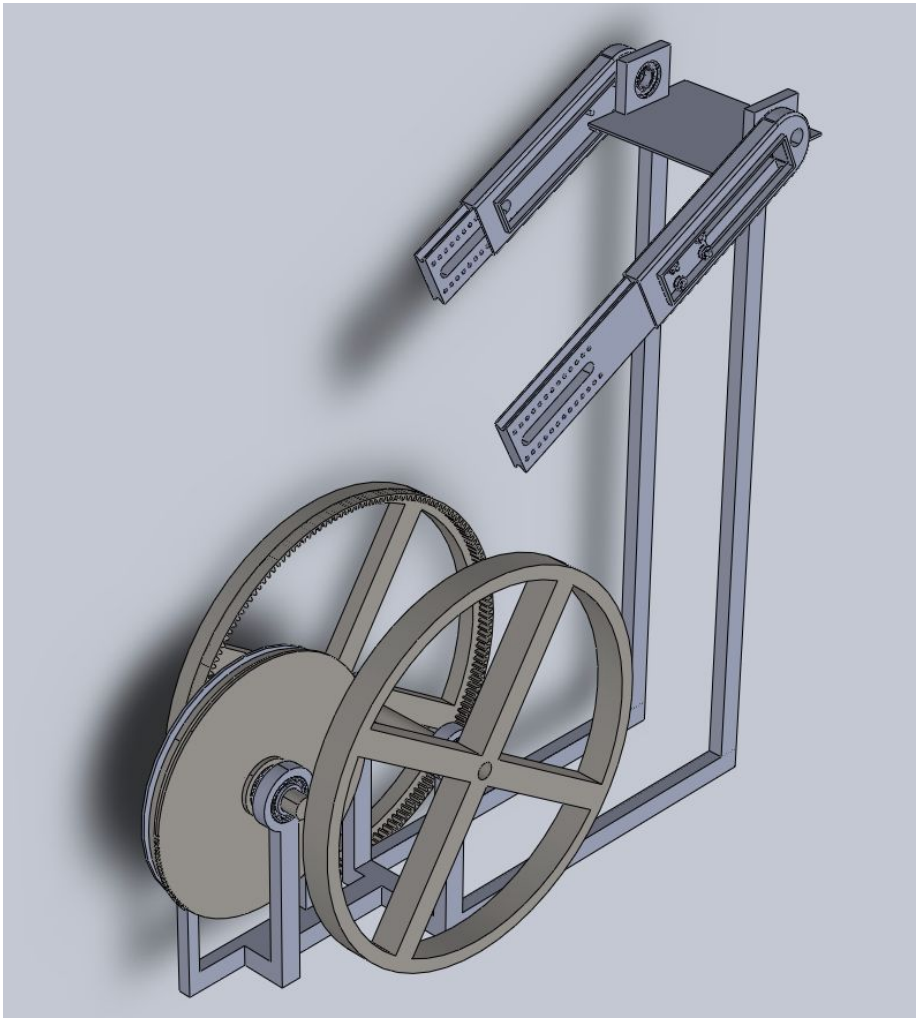
Parameter	Target Spec	Selected Approach
Exercise modes	2, seated	Cycling and flexion/extension; position-based switching
Cycling resistance	Stepwise adjustable	Eddy current: aluminium flywheel between 2 discs of 8 permanent magnets each; toothed moving plate sets the gap
Flexion/extension resistance	Continuous, adjustable	DC motor with planetary gearbox
Range of motion	User-configurable at setup	Set in software at the start of a session
Footprint / weight	[600] × [500] × [800] cm / [15] kg *(rough)*	Compact, floor-anchored
Max user weight	100 kg	Design target
Power failure	Safe arm return	Supercapacitor discharge
Feedback	Reps, ROM, cadence	Encoders + display and voice guidance

Benchmarking and Innovation

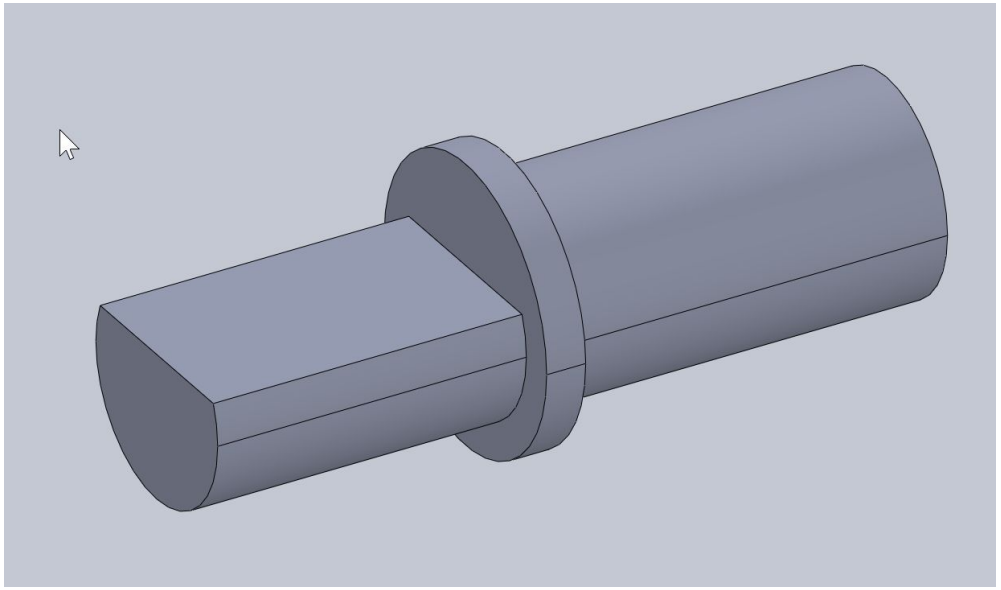
- 15 existing devices and patents were benchmarked, including Biodex System 4 Pro, Lode Corival, Keiser M3i, Wattbike, Peloton Bike+, Monark 894E and MRKneeExo (UFES).
- Existing machines handle cycling **or** isokinetic strengthening separately. We found no existing product that combines both in one compact, floor-anchored home device.
- **Ideas adopted:**
  - Eddy-current braking (Keiser, Biodex)
  - Encoder-based ROM tracking (Lode, Biodex)
  - Configurable ROM (Biodex)
  - Planetary gearbox for torque (Wittenstein)
  - Supercapacitor power-failure backup (US7331426B2)
- **Rejected:** linear actuator, cam and solenoid latches, because they add parts and complexity. Repositioning is simpler and more reliable.

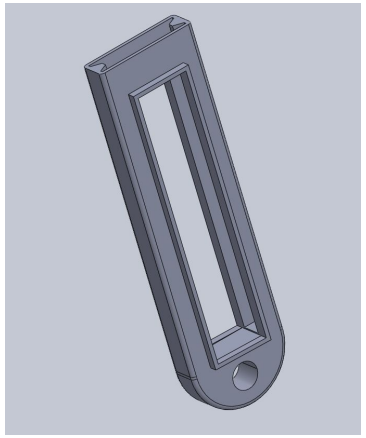
CAD Design & Mechanical Specs

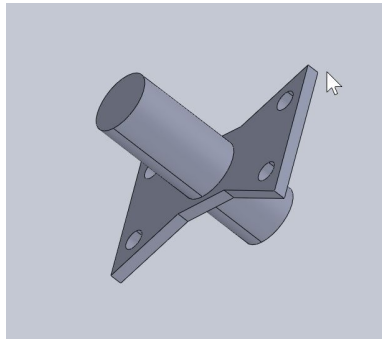


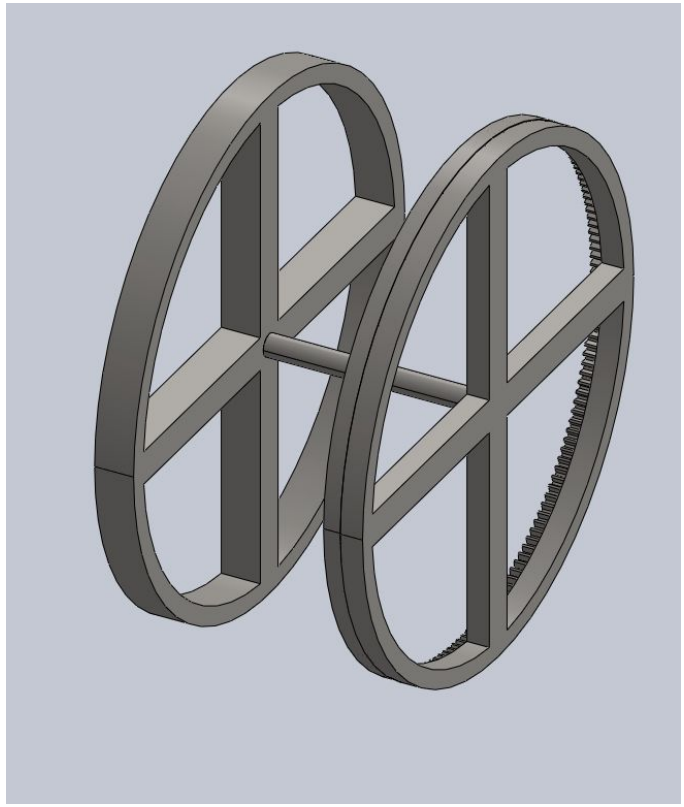


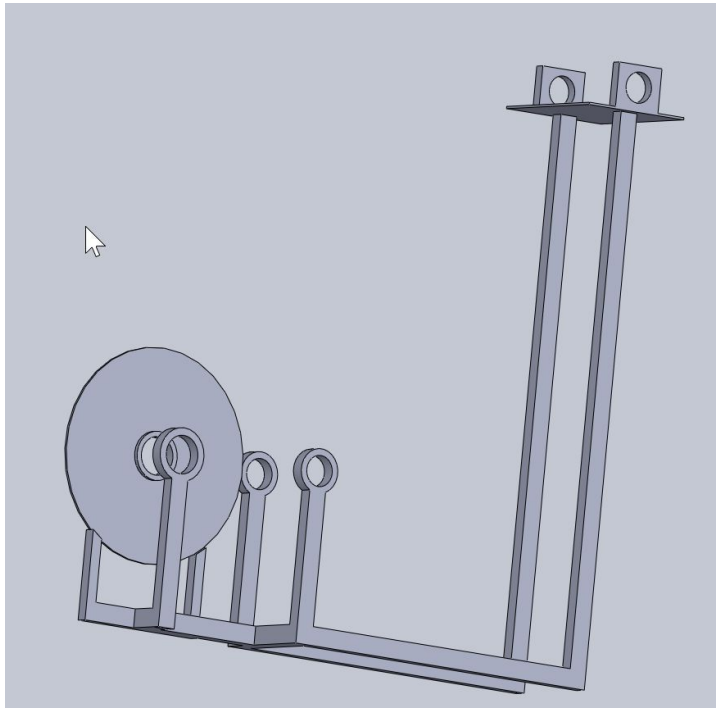
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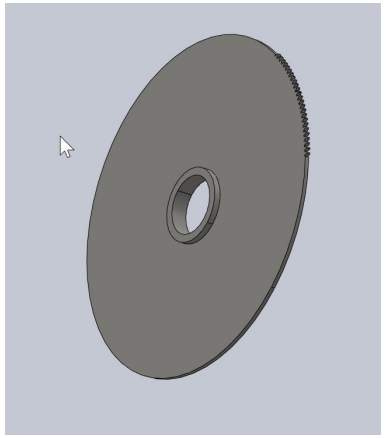




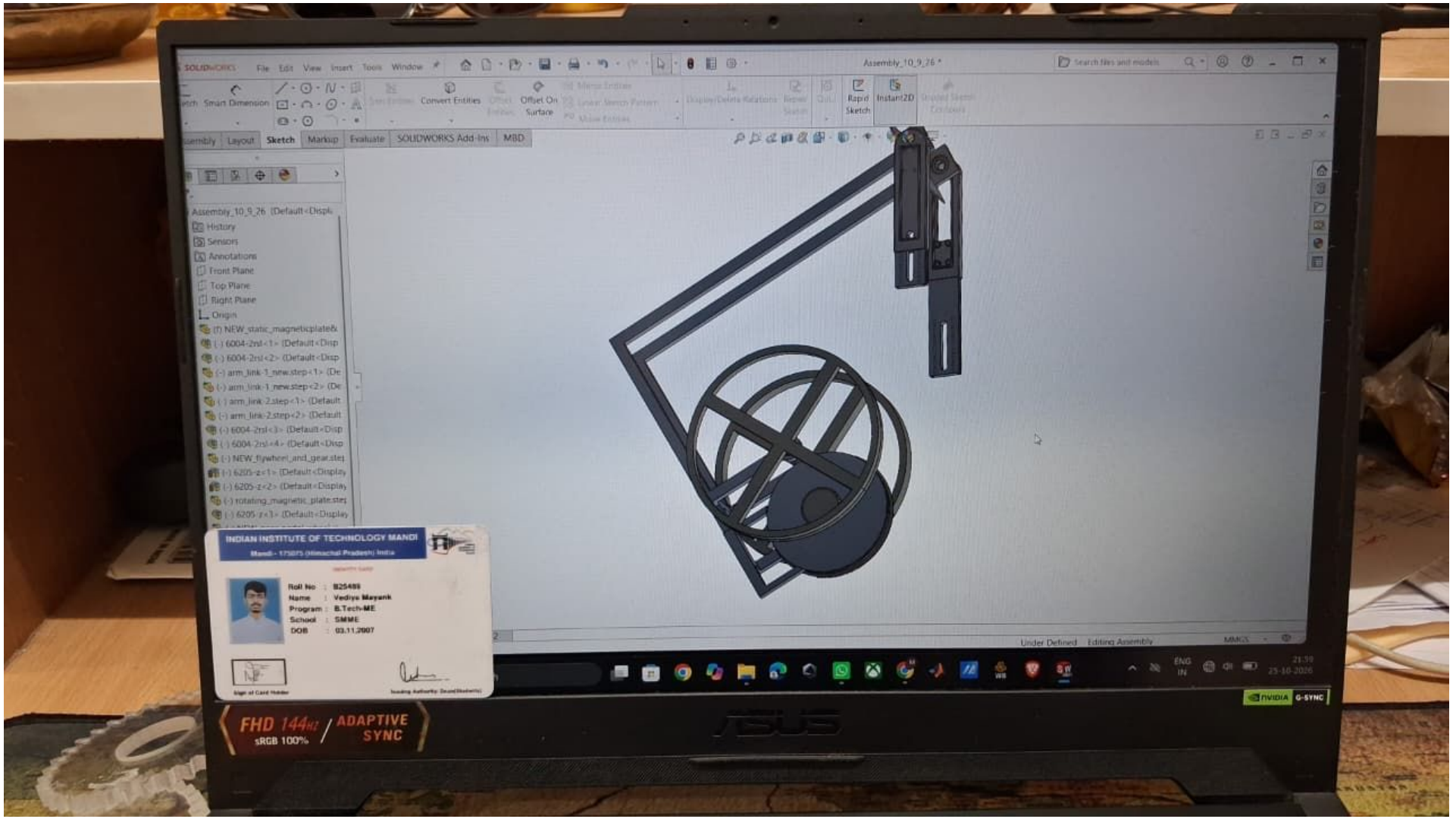




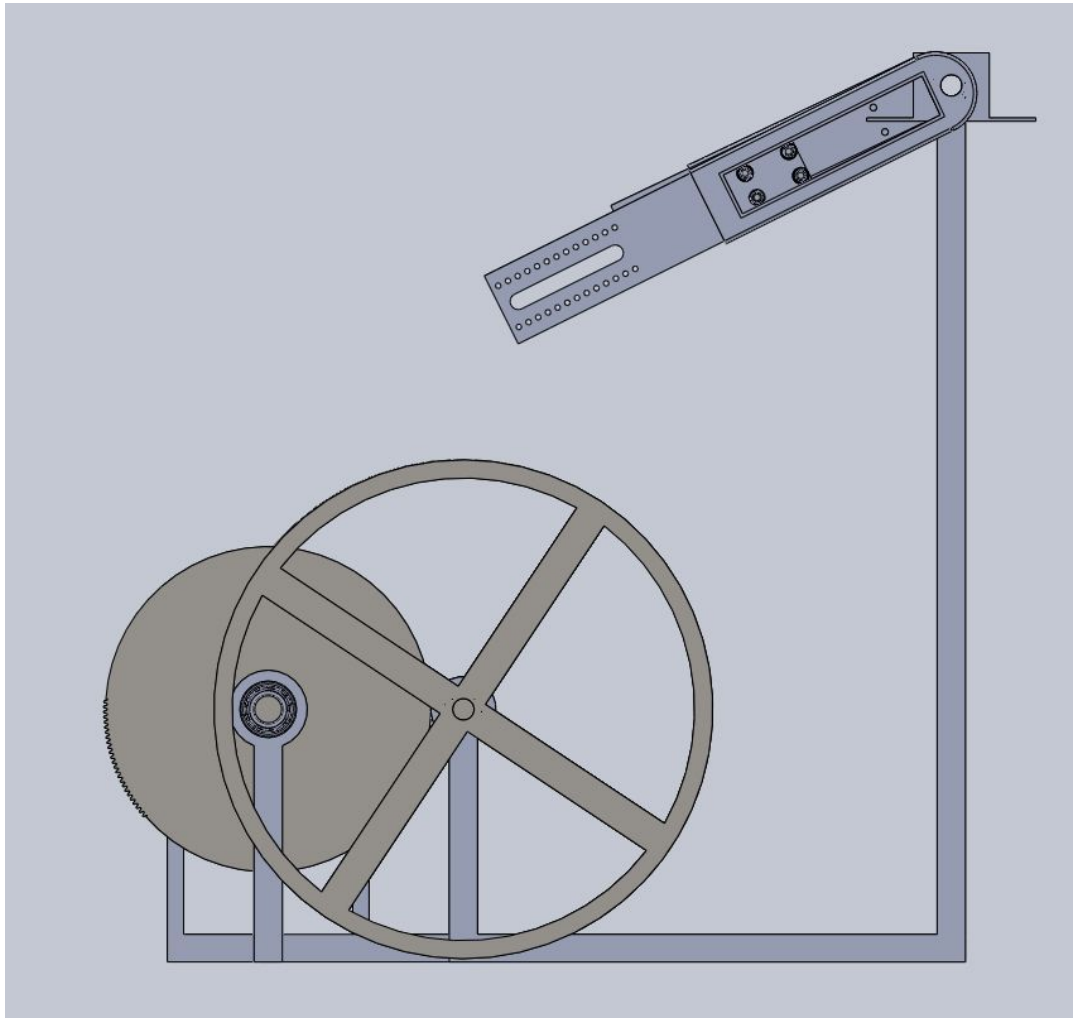


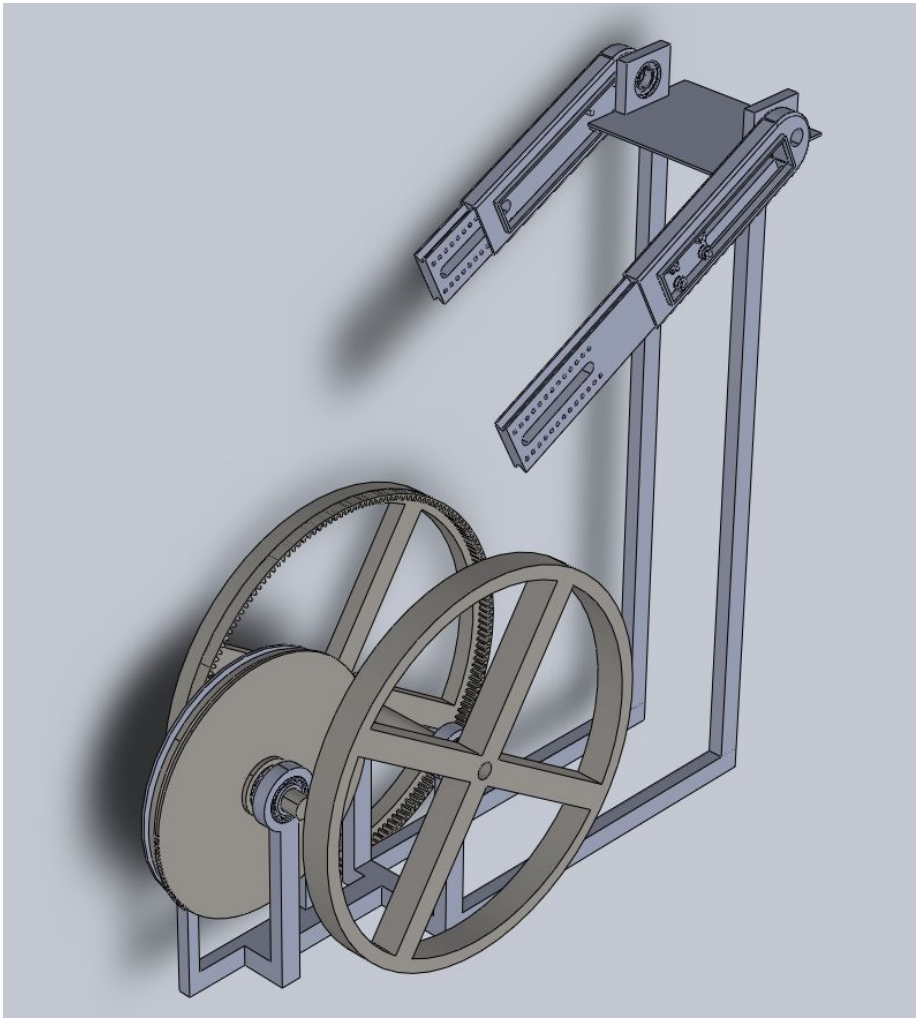


CAD Design & Mechanical Specs



CAD Design & Mechanical Specs





CAD Design & Mechanical Specs

3D CAD

- Modelled in SolidWorks as separate components (flywheel, pedal assembly, gear, flexion-extension arm links, magnetic plates), then integrated into one assembly.

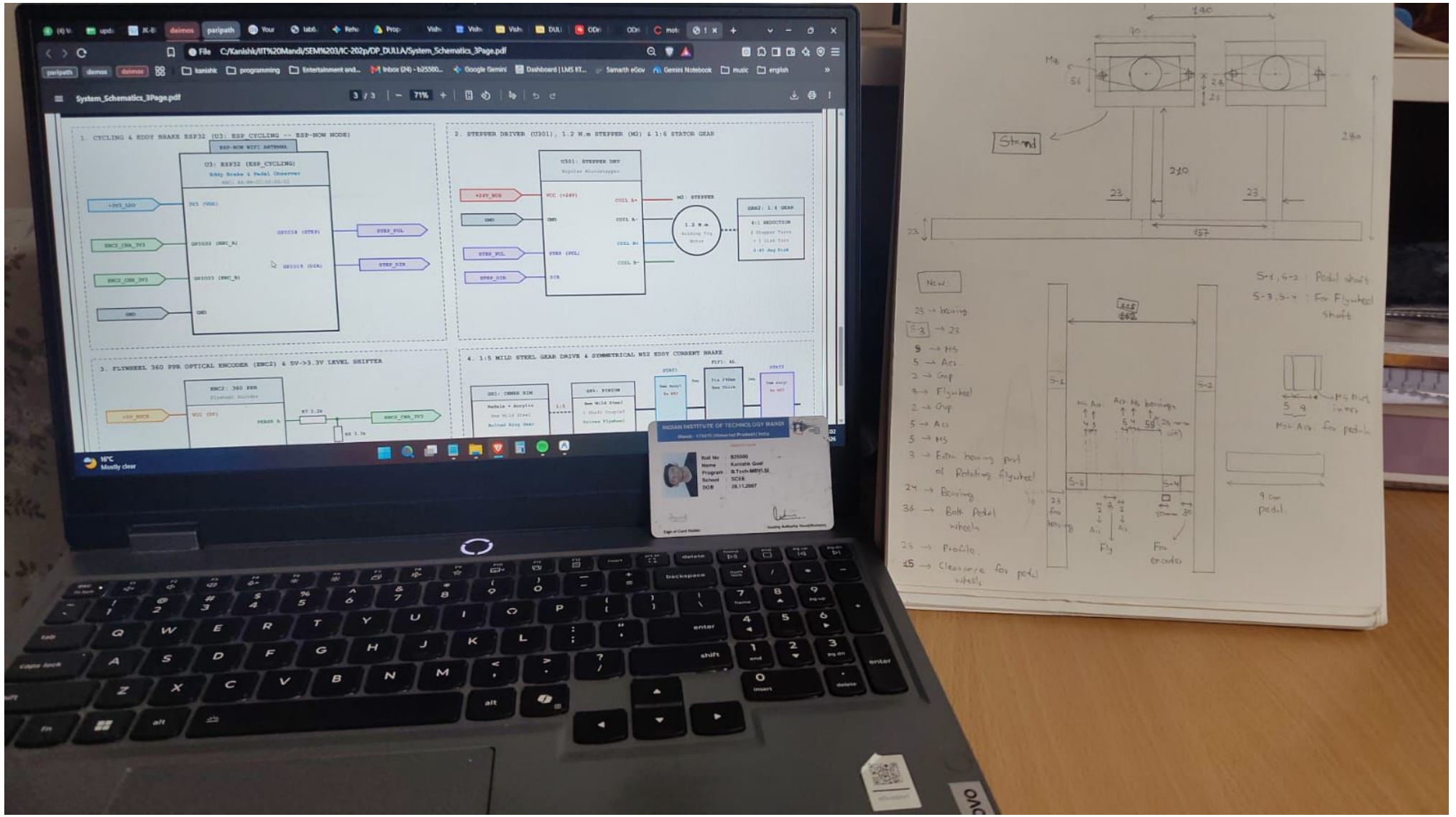
Material Selection

- Flywheel: Aluminium (conductive, so it supports eddy-current braking)
- Magnetic plates: 2 discs with 8 permanent magnets each
- Frame and arm links: [material, e.g. Aluminium 6061 / mild steel] chosen for [strength / weight / cost]. *Fallback: "To be finalised in Stage 3 based on stress analysis"*
- Pedals: wide platform with spring-loaded retaining flaps

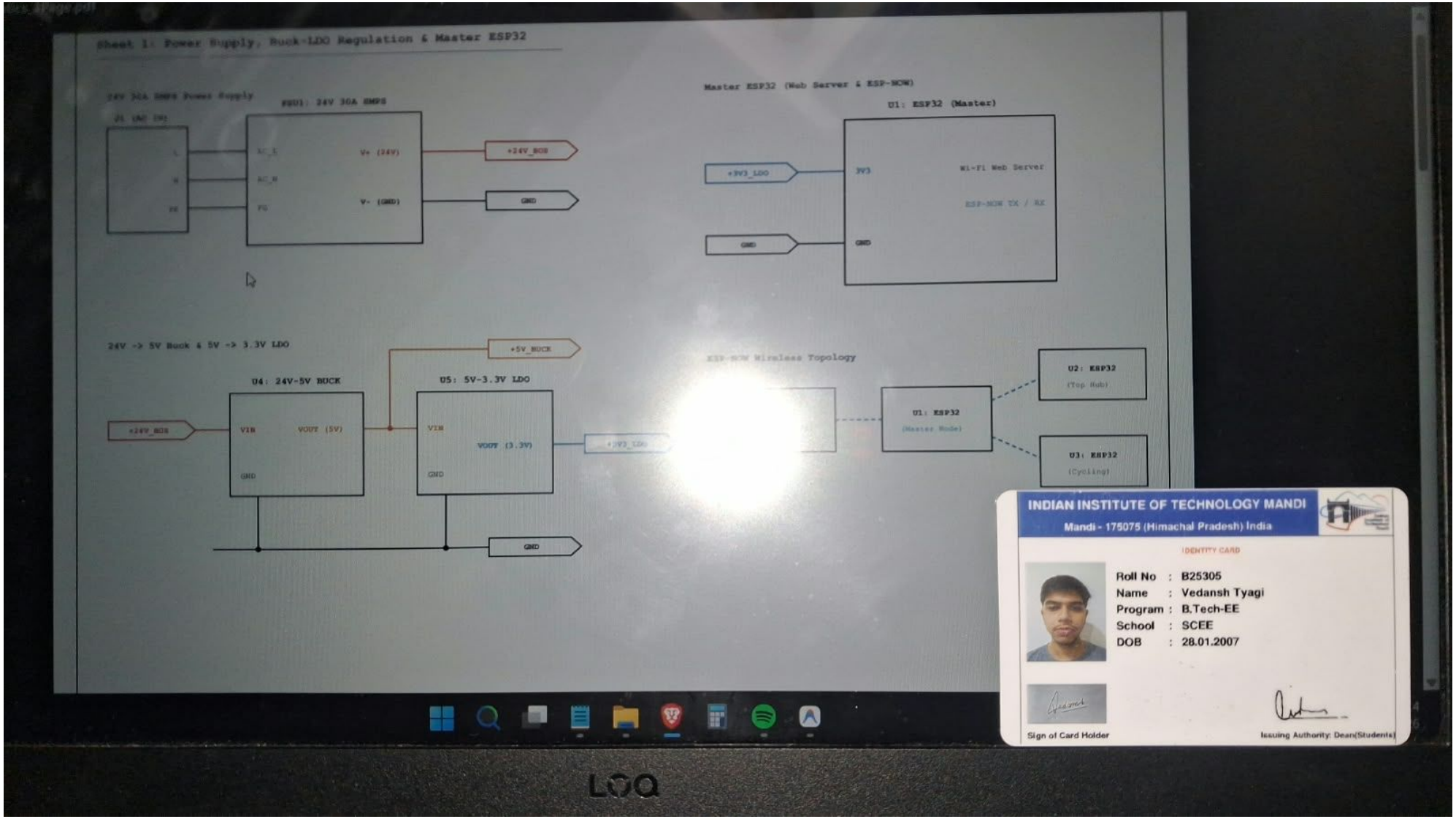
Design Iteration

- Mode-switching latch (linear actuator / cam / solenoid) replaced by repositioning, reducing parts and complexity.
- Elbow-like joint added to the flexion-extension arm so it tucks clear during cycling.
- Leg gap widened in flexion/extension mode to avoid contact with protruding pedals.

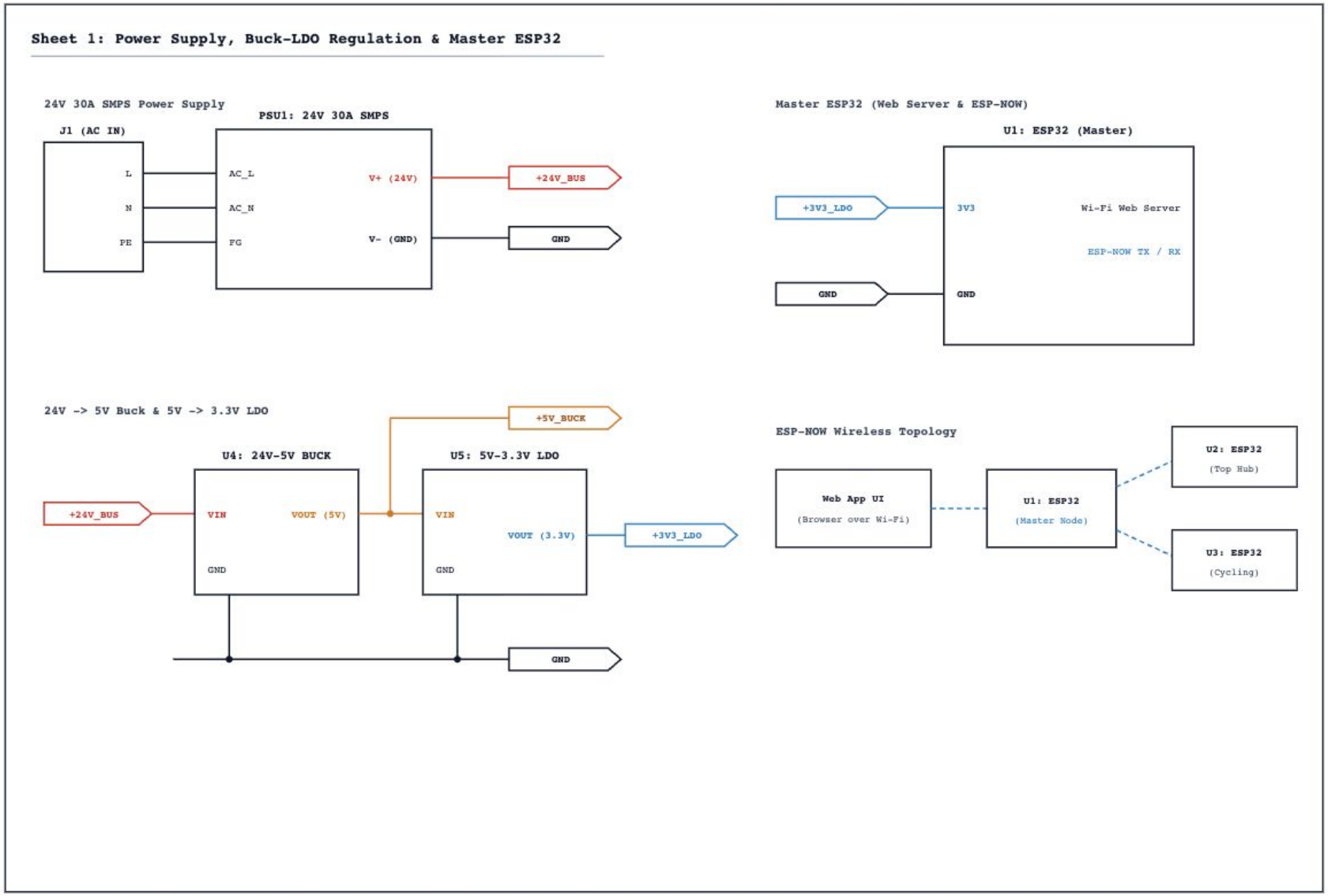
Electronics Schematics & BOM



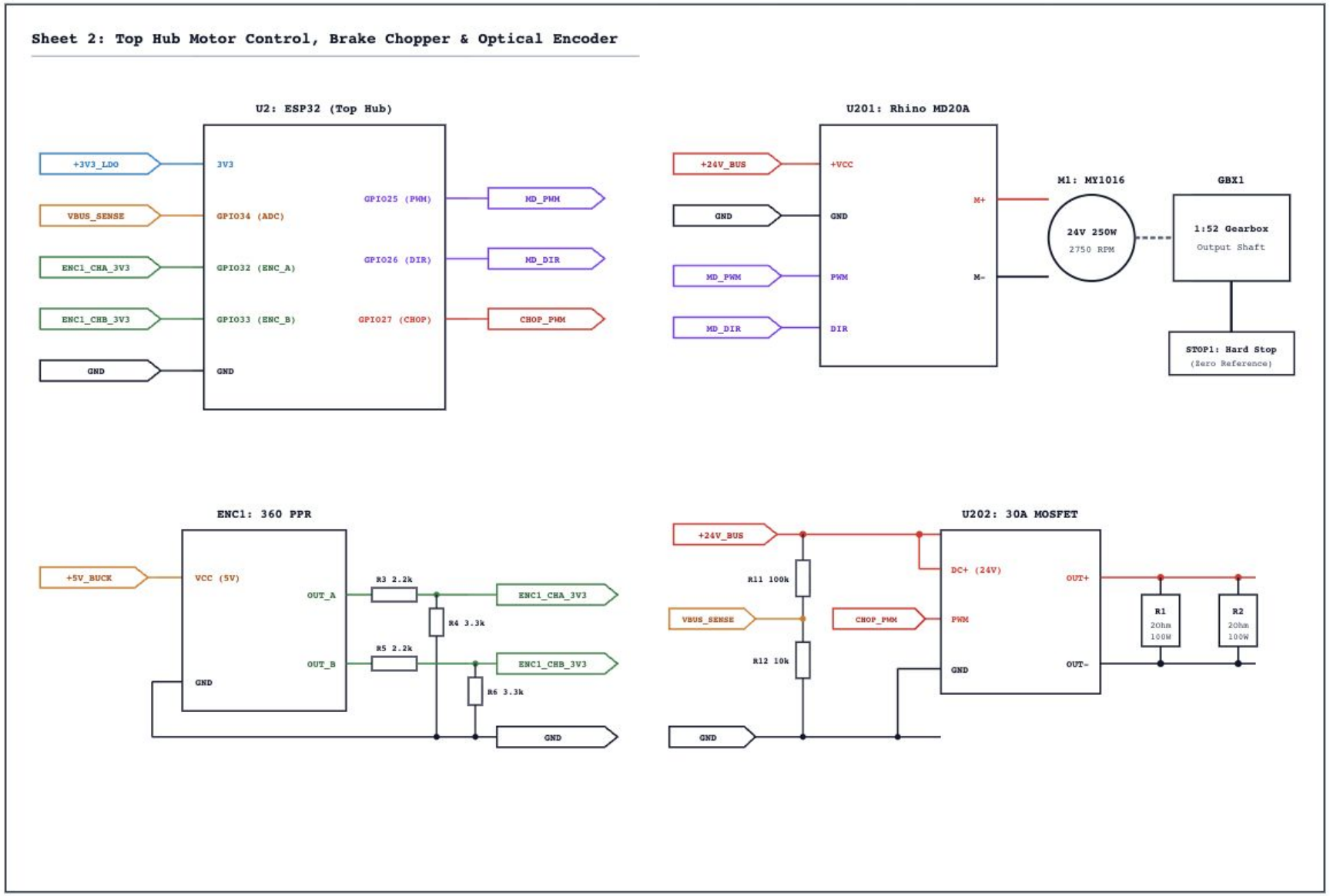
Electronics Schematics & BOM



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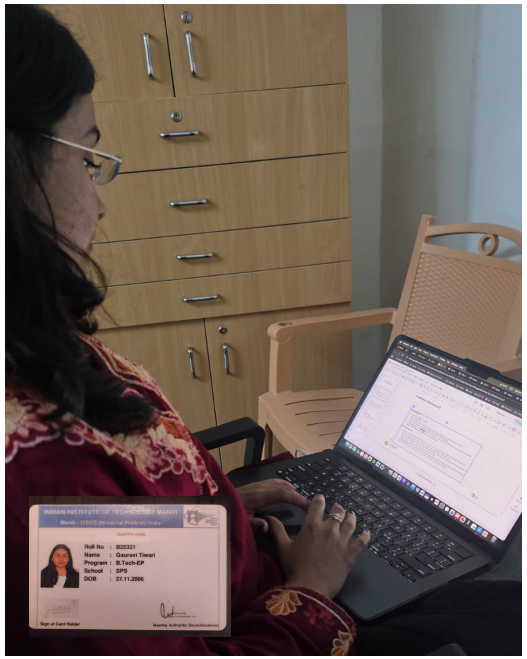
Circuit Overview (preliminary)

- Power: mains → SMPS → controller and motor driver, with a supercapacitor bank on the backup path
- Control: controller → motor driver → DC motor; encoder inputs for ROM, reps and cadence
- Safety: force/jerk sensor input *(planned)*; optional NTC thermistor on the flywheel *(considered)*

REF DES	COMPONENT	QTY	SPECIFICATION / DESCRIPTION
PSU1	24V 30A SMPS Power Supply	1	24V DC, 30A (720W) Switching Power Supply (Sheet 1)
U4	24V-to-5V Buck Converter	1	Steps down 24V bus to 5V for Encoders & LDO input (Sheet 1)
U5	5V-to-3.3V LDO Regulator	1	Linear regulator converting 5V to 3.3V for ESP32s (Sheet 1)
U1, U2, U3	ESP32 Microcontroller	3	U1: Master Web Server, U2: Top Hub, U3: Cycling (ESP-NOW)
U201	Rhino MD20Amp Motor Driver	1	6V-30V, 20A Single-Channel DC Motor Driver (Sheet 2)
M1	E-Bike DC Motor MY1016	1	24V DC, 250W, 2750 RPM Permanent Magnet DC Motor (Sheet 2)
GBX1, STOP1	1:52 Gearbox & Hard Stop	1	1:52 reduction gearbox on M1 + mechanical zero stop (Sheet 2)
U202	30A Heatbed MOSFET Module	1	3D Printer 30A MOSFET module for PWM brake chopper (Sheet 2)
R1, R2	2 Ohm 100W Braking Resistors	2	Ceramic aluminum-shell resistors wired in parallel (Sheet 2)
ENC1, ENC2	360 PPR Optical Encoder	2	2-phase incremental rotary encoders, 5V supply (Sheets 2, 3)
R3 - R10	Encoder Level-Shift Resistors	8	4x 2.2kOhm + 4x 3.3kOhm (5V to 3.3V divider for ENC1, ENC2)
R11, R12	Bus Voltage Sense Resistors	2	100kOhm & 10kOhm (24V to 3.3V divider for brake chopper ADC)
U301, M2	1.2 N.m Stepper & Driver	1	1.2 N.m holding torque stepper + driver + 1:6 gear GBX2 (Sheet 3)
GR1, GP1	1:5 Mild Steel Gear Set	1	4mm mild steel inner rim gear (on pedal acrylic) & pinion (1:5)
FLY1	Aluminum Flywheel Disk	1	290mm diameter, 8mm thick solid aluminum disk (Sheet 3)
STAT1, STAT2	Symmetrical N52 Magnet Disks	2	5mm acrylic sheets, 2mm air gap, 8x (30x10x5mm) N52 magnets/side

Maker Journey





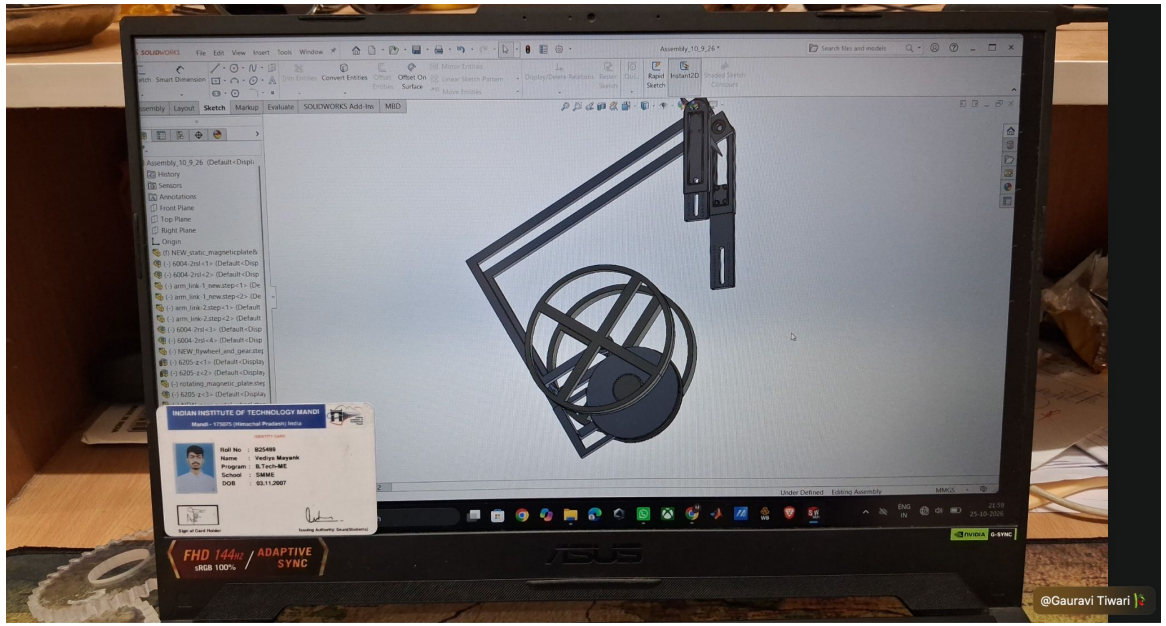




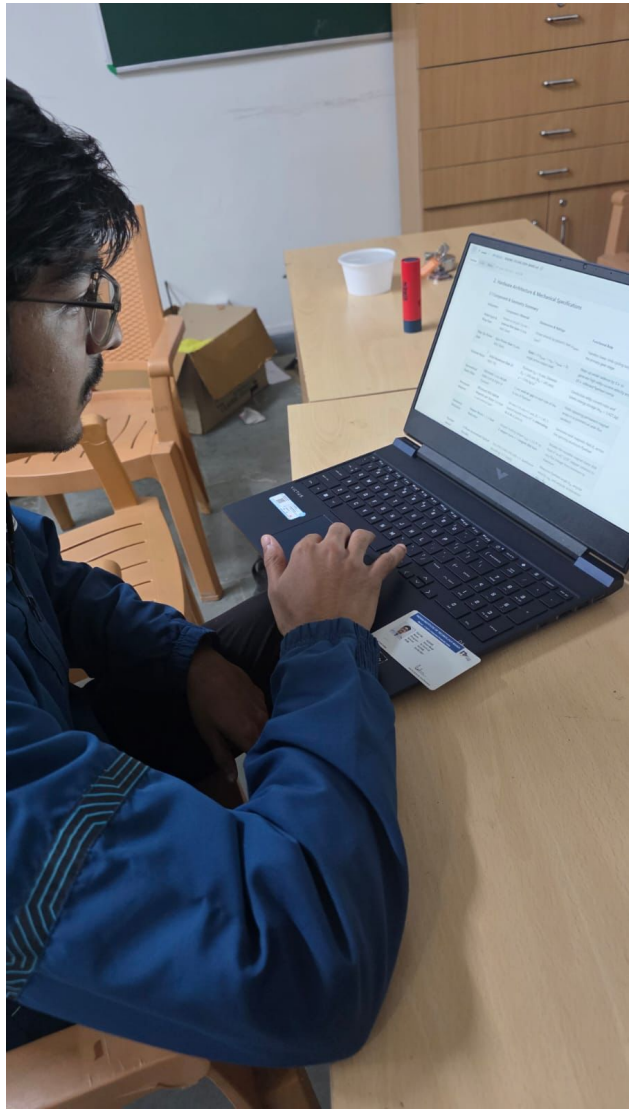
Maker Journey







Maker Journey





Maker Journey

Stage 1 to Stage 2 Engineering Iterations

- **Concept phase:** Discussions with mentor Dia Bhandari on integrating cycling and flexion-extension; compared options for mode switching, resistance, pedals, adjustability and safety.
- **Design pivot:** Dropped the latch-based mode switch in favour of repositioning; chose a DC motor for flexion/extension resistance and eddy-current braking for cycling.
- **CAD milestone:** Complete SolidWorks assembly of the flywheel, gear-pedal wheel, pedal attachment, arm links and magnetic plates
- **Benchmarking:** 15 existing devices and patents reviewed.

Verified Drive Repository: https://drive.google.com/drive/folders/1woBixmS_sJiRxzflLFP72UtD1tfvnQPf?usp=sharing, containing SolidWorks files and schematic

Github Repository: https://github.com/Kanishk-Goel-19/DP_DULLA

Mentor Guidance: Guided by Dr. Gajendra Singh and Miss Dia Bhandari, IIT Mandi.

Safety & Hazard Mitigation

Physical Fail-safes

- Floor-anchored base with seated, adjustable chair
- Spring-loaded wide pedals keep the foot in place
- Arm tucks away during cycling; wider leg gap in flexion/extension mode
- Supercapacitor returns the arm to a safe position on power failure

Standards Compliance *(design references; not yet tested or certified)*

- ISO 20957: stationary training equipment
- IEC 60601-1 and 60601-1-11: medical electrical equipment, home use
- ISO 14971: risk management for medical devices

Failure Mode	Potential Hazard	Mitigation
Power cut in flexion/extension	Arm stuck with leg bent	Supercapacitor drives arm to a safe position
Muscle spasm / sudden force change	Joint strain	Force/jerk sensor triggers motor stop *(planned)*
Encoder or sensor fault	Wrong ROM or rep count	Software range limits; stop on invalid reading
Foot slips off pedal	Fall or leg injury	Wide pedals with spring-loaded retaining flaps
Flywheel heating	Resistance drift	NTC temperature compensation *(considered)*
Leg hits protruding part	Bruising	Tucked arm; widened leg gap

Roadmap

4-Month Fabrication Plan

- **Month 1 (Nov 2026):** Finalise CAD; run FEA and motion simulation; finalise BOM and place orders.
- **Month 2 (Dec 2026):** Machine frame, flywheel and magnetic plates at the IIT Mandi workshop; bench-test motor, driver and encoders.
- **Month 3 (Jan 2027):** Assemble the mechanical structure; integrate electronics and firmware; add the supercapacitor safety circuit.
- **Month 4 (Feb 2027):** Test range of motion, resistance steps, mode switching and stability; tune and document.

Stage 3 Target: A working prototype with both modes, adjustable resistance and basic feedback, ready for functional testing and the live demo.

Technical Challenges & Risks

Engineering Bottlenecks

- **Mode interference:** pedals, arm and legs must not collide in either mode. This led to the elbow joint and wider leg gap.
- **Torque vs size:** the motor must give enough flexion/extension torque without making the device bulky.
- **Eddy-current drift:** resistance drops as the flywheel heats, and is hard to hold at low cadence.
- **Spasm detection:** setting a threshold that catches real spasms without false stops.
- **Cost vs safety:** staying affordable while meeting safety needs.
- **Validation status:** no simulation or physical testing yet, so all values are design targets.

Potential Solutions & Risk Mitigation

Risk Mitigation

- **Heat drift:** NTC sensor on the flywheel to compensate resistance.
- **Low-cadence resistance:** variable gear ratio between pedal axle and flywheel (from existing patents).
- **Torque:** planetary gearbox; change motor or gear ratio if needed.
- **Interference:** adjust the arm's elbow joint and leg gap after motion simulation (one-time screw adjustments).
- **Switching fallback:** if repositioning proves awkward, fall back to a solenoid or cam latch (both already evaluated).
- **Sensor fault:** software limits on ROM and torque if a sensor fails.

References & Sources

- SolidWorks CAD screenshots: *Own work, SolidWorks*
- Before/After flowchart: *[AI-generated]*
- Team photos: *Own photos, IIT Mandi*
- Block diagram: *Own work*
- US Patent 3,947,140 (expired, public domain)
- US Patent US7331426B2 (expired)
- US Patents 4,678,182 and 9,999,818 (variable gear ratio, considered)
- Benchmark sources: Biodex, Lode, Keiser, Wattbike, Peloton, Monark, Placid Industries, Ledex, Wittenstein alpha, UFES MRKneeExo